CRTFE V-2

INTEGRATED ENGINEERING, WIRING AND ASSEMBLY MANUAL

Revision P2 - Ground-Research Electrical Architecture

Includes the P2 eight-sheet electrical schematic set and the retained V5 P0 energy-sled mechanical/fastener baseline.

CONCEPT ONLY - NOT A FABRICATION DRAWING - NOT FLIGHT READY - NO ENERGIZED WORK AUTHORIZED

Document	Controlled value
Configuration	V0.3 / V-2 G2 ground articles plus V5 target-vehicle interface
Revision	P2 - 2026-08-24
Maturity	DRAFT concept-level integration
Authority	No flight, fabrication, live-cell, HV-source construction or energized-test authority

1. Revision P2 change summary

P2 adds the missing connection architecture and makes the trust and energy boundaries explicit.

- Eight-sheet blue-pencil/graph-paper electrical schematic set.
- V0.3 system interconnect and hardwired safety-chain concept.
- G2B plasma-off polyphase stator validation one-line.
- G2D combined HTS-bias/stator wiring and independent quench boundary.
- V5 energy-sled functional single-line and ARC power/data/safety separation.
- Cable-class, interface, inspection, assembly and staged-commissioning schedules.
- Existing P0 structural, shield, fastener and inert-mockup manual retained as a controlled appendix.

P2 does not invent values that the project has not earned. Power conductor sizes, HV ratings, contactor/fuse ratings, coil turns/current, quench dump values, creepage/clearance and live test limits remain TBD pending selected hardware and professional engineering.

Document organization

Section	Contents
Part I	P2 electrical integration, wiring, assembly, inspection and commissioning
Appendix A	Retained V5 P0 energy-sled structural/fastener/assembly manual
Appendix B	P2 electrical wiring schematic set E-001 through E-601

2. System purpose in plain language

The first job is not to build an aircraft. The first job is to test one hard question: can moving air be made electrically conductive enough, without unacceptable power or heat, to justify the next experiment?

Article	Simple purpose	What it can prove
V0.3	Move air through a small duct; measure conductivity, power, heat, decay and flow.	Whether the assumed conductive-air state exists in the selected regime.
G2B	Power the bare traveling-wave stator with no plasma; map field and coil power.	Whether the finite-coil model predicts the real winding.
G2D	Combine the HTS bias magnet and stator with no plasma.	Whether they coexist without unacceptable loss, heating or margin loss.
V5 interface	Preserve a target-vehicle packaging and safety boundary.	Nothing about flight until every prior gate is passed.

The source files use CRFTE and CRTFE. No approved letter-by-letter expansion is present, so this manual does not invent one. The controlled plain-language name is CRTFE Experimental Aerospace Research Project.

3. Electrical architecture and responsibility

Layer	Function	Responsible authority
Facility power	Branch protection, LOTO, grounding and safe work condition	Host laboratory / qualified electrical personnel
Hardwired safety	E-stop, guards, protective trips, manual reset	Qualified safety/electrical integrator
Test power	Blower, plasma subsystem, stator source and HTS supply	Equipment owner and responsible engineer
Instrumentation	Calibrated V/I/Z, field, temperature, velocity and time	Test engineer
DAQ	Synchronized raw data and configuration records	Test engineer; no protective authority
ARC/T.A.R.	Evidence retrieval, analysis and non-executable planning	Authenticated human review; no actuator authority

The hardwired safety chain is independent of ARC, T.A.R., the DAQ computer, ordinary software and network availability. Opening a guarded energized compartment must remove the applicable hazardous-energy enable using a validated architecture. Manual reset and no automatic restart are required.

Safety references

- OSHA 29 CFR 1910.147: energy isolation and lockout/tagout.
- OSHA 29 CFR 1910.333: electrical safe work practices.
- OSHA 29 CFR 1910.306: interlock precedent for access panels on specific equipment.
- NFPA 70E: electrical safety program and shock/arc-flash risk framework.

4. Cable-class schedule

Class	Service	P2 planning rule	Release
C0	Protective earth/bonding	Dedicated studs; no signal-current return	TBD by fault/code study
C1	24 VDC safety/control	Twisted stranded copper; separately fused	18 AWG bench candidate only
C2	Analog sensors	Shielded twisted pair	22-24 AWG bench candidate only
C3	Thermocouple	Correct alloy extension wire	24 AWG bench candidate only
C4	Data/time	Fiber preferred across noisy boundaries	By network/EMC design
C5	Stator A/B/C	Symmetric phase routing	TBD after frequency/current/thermal study
C6	HTS DC/leads	Vendor-qualified cryogenic interfaces	TBD by magnet design
C7/C8	Propulsion HV/special disconnect	Segregated, touch-safe, guarded	Not released in P2

Small-wire gauges are bench candidates only. The responsible engineer must verify branch protection, voltage drop, temperature, bundle derating, flexing, connector compatibility and the host facility's rules before use.

5. Interface and connector schedule

ID	Interface	Mandatory feature
J-V03-CTRL	V0.3 control box	Keyed low-voltage; control/status only
J-V03-DAQ	V0.3 instruments	Shielded pairs; no hazardous-energy conductors
J-V03-HV	Enclosed plasma subsystem	Lab-owned, touch-safe, interlocked; pinout not released
J-G2-PH	G2 phase interface	A/B/C identity; protective bonding as designed
J-G2-SENSE	G2 measurement	Finger-safe lab-selected V/I/temperature interface
J-HTS-DC	HTS supply/current leads	Vendor/engineer controlled; no P2 pinout
J-HTS-QD	Quench/dump	Independent protection; ARC gets status only
J-SLED-HV	Energy-sled propulsion HV	Touch-safe, keyed, HVIL candidate; rating TBD
J-SLED-LV	Energy-sled essential LV	BMS/IMD/status and shutdown request
J-ARC-A/B	ARC power/data	Isolated paths; no direct actuator pins

Connector manufacturer, family, shell, insert, pin assignment, sealing, mating cycles and keying remain configuration-controlled selections. Never infer a pinout from a concept drawing.

6. Wire identification and workmanship

- Assign every conductor a unique wire ID and show the same ID on schematic, wire list and both physical ends.
- Record from/to equipment, connector and pin, service class, conductor/cable part number, gauge, length, shield rule, protection and inspection status.
- Use controlled crimp tooling matched to the terminal and conductor. Record tool identification and inspection status.
- Provide strain relief, bend-radius control, chafe protection, environmental sealing and a service loop where removal requires it.
- Keep protective bonding separate from signal return and shield termination.
- Do not splice hazardous-energy wiring unless the released design and facility procedure explicitly permit it.
- Inspect 100 percent of safety-chain, HTS-protection and propulsion-HV terminations.

NASA-STD-8739.4 is used as a workmanship reference for cable and harness assemblies. FAA AC 43.13-1B may inform aviation workmanship only where applicable and legally permitted. Neither reference is a substitute for the selected component manufacturer's instructions or a project-specific engineering release.

7. Grounding, bonding, shielding and field compatibility

Topic	P2 rule	Verification
Protective bond	Dedicated low-impedance path; no hinge-only bond	Low-resistance measurement and visual inspection
Signal reference	Defined by instrument/DAQ architecture	Noise and common-mode test
Cable shield	Terminate per EMC plan; avoid improvised pigtails	Continuity and transfer-impedance review
Conductive enclosure	Segment/slot/insulate where analysis requires	G2B/G2C field-loss and heating comparison
Carbon/composite	Control galvanic and bonding interfaces	Coupon and environmental test
Crossings	Documented locations, preferably near 90 degrees	Harness-zone inspection

A shield that protects against impact or fire can become a conductive shorted turn near a changing magnetic field. G2C must measure field distortion, induced loss and heating with the installed shield, seams and representative fasteners.

8. Assembly sequence - drawings and enclosure

- Freeze configuration ID, schematic revision, test plan and approved equipment list.
- Create the complete wire list and connector schedule before cutting wire.
- Complete fault-current, ampacity, insulation, creepage/clearance, stored-energy and EMC analyses before hazardous-energy release.
- Install enclosure, DIN rail, terminal blocks, protective-earth bar and labeled cable-entry plates.
- Install the hardwired safety components before ordinary control or DAQ hardware.
- Install separate routing zones for C0/C1, C2/C3/C4 and C5/C6/C7.
- Bond the enclosure and mounting panels first; document the hardware stack and measured bond.
- Build and label low-voltage harnesses to the controlled cut list.
- Install hazardous-energy harnesses only after engineering and host-laboratory release.

No work beneath an unsupported energy sled. No live cells, charged capacitors, HTS stored energy or plasma-source connection during mechanical fit check.

9. De-energized inspection and verification

Step	Evidence
Point-to-point	Every net agrees with schematic/wire list; checker signs
Isolation	No unintended continuity among power, control, sensor, shield and chassis
Bonding	Recorded low-resistance result using approved method
Connector	Keying, polarization, backshell, strain relief and labels correct
Harness	Separation, bend radius, clamps, chafe and service loops correct
Safety logic	E-stop/guard/manual-reset tested with low-energy supply
Restart	Power cycle cannot cause automatic hazardous-energy restart
Data	Channel map, calibration IDs, timebase and raw-data path verified

Insulation-resistance or hipot testing can damage electronics or create hazardous stored charge. The responsible engineer must define the method, disconnected components, test voltage, discharge and acceptance criteria. P2 does not invent them.

10. Staged commissioning

Stage	Allowed activity	Stop condition
C0	Mechanical fit and labels; all sources disconnected	Any interference, chafe or connector error
C1	Low-energy safety-chain test	Any trip/reset/feedback discrepancy
C2	DAQ, sensors and time synchronization	Missing calibration or time integrity
C3	Blower and V0.3 plasma-OFF baselines	Unstable flow, vibration, leak or data gap
C4	G2B stator at lowest approved energy; no plasma	Unexpected imbalance, heating or field
C5	G2C passive hardware installed	Shorted-turn signature or unexplained loss
C6	G2D HTS + stator; no plasma	Margin loss, quench anomaly or excessive AC loss
C7	Plasma-loaded test under host procedure	Arc, excessive heat, interlock or data anomaly

Advancement is evidence-gated. Passing a wiring inspection does not validate propulsion, and passing a field-map test does not validate thrust.

11. V0.3 operating record

Every plasma-ON run must cite a matching plasma-OFF baseline at the same nominal 10, 20 or 30 m/s flow point. The minimum run record includes:

- Run, configuration, operator, facility, date/time and calibration IDs.
- Complete 5 x 5 XY velocity plane and axial location.
- T1/T2 and ambient temperature/pressure/humidity where available.
- Z20, Z50 and Z80 complex impedance records.
- Plasma-source electrical input from the qualified facility instrumentation.
- Synchronized video and discharge classification: diffuse, filamentary, thermal/arc or uncertain.
- Raw data preserved separately from derived conductivity and model output.
- Stop/pivot decision against the pre-committed conductivity, power, heat and morphology gates.

No V0.3 data become vehicle-scale proof. The 0.01 m^2 duct is approximately 60 times smaller than the earlier 0.60 m^2 module model and is a screening article only.

12. G2 electromagnetic validation record

- Mesh-converged finite-coil/circuit model with reproducible inputs and hashes.
- Bare-coil vector map: at least five axial stations and 5 x 5 cross-section grid per station.
- Field amplitude and phase, complete coil impedance matrix, real/reactive/apparent power, power factor, harmonics and temperature rise.
- Installed passive hardware repeat with duct, cryostat, supports, shield, seams and representative fasteners.
- Combined HTS/stator repeat with temperature margin, AC loss, stored energy and protection response.
- Pre-registered acceptance limits derived from calibrated uncertainty and design sensitivity.

Preliminary comparison objectives in the project record are <=5 percent field-amplitude RMS error, <=10 percent maximum local field error, <=5 degrees phase error and <=10 percent real-power error. They are not final released acceptance criteria.

13. Energy sled and ARC interface

The V5 energy sled remains a target-vehicle packaging and safety study: 1450 x 1050 x 340 mm maximum envelope and 120 kg battery-system allowance. Chemistry, bus voltage, cell count, fault current and live hardware are not selected.

Function	Required order/boundary
HV path	Segmented modules -> service disconnect -> main fuse -> K+/K- -> precharge/discharge -> DC link -> isolated branches
Protection	BMS, IMD, fuse, contactor feedback and service disconnect remain independent of ARC
Low voltage	Isolated essential supply supports recorder and shutdown functions
ARC	Receives approved status and may submit non-executable intent; no contactor-drive or actuator pins
Shield	Impact/fire protection must also pass electromagnetic loss/heating tests

14. Hold points and release evidence

Hold point	Evidence	Authority
HP-E1	Approved schematic, wire list, component data	Responsible electrical engineer
HP-E2	Bond/enclosure inspection	Engineer / facility safety
HP-E3	Netlist verification and insulation test plan	Independent checker
HP-E4	Low-energy E-stop/interlock/manual-reset test	Facility safety representative
HP-E5	DAQ calibration and time synchronization	Test engineer
HP-E6	V0.3 and G2 plasma-OFF baselines	Test director
HP-E7	HTS quench/dump validation	Magnet/cryogenic engineer
HP-E8	Energized-test readiness review	Host laboratory authority

A signature at one hold point does not waive another discipline's authority. Safety, facility and test authorities remain independently veto-capable.

15. Stop-work and discrepancy control

- Unexpected continuity, insulation or protective-bond result.
- Missing or incorrect conductor, connector or equipment label.
- Guard opening does not remove the relevant enable.
- Automatic restart after control-power restoration.
- Welded/inconsistent contactor feedback or loss of protective bond.
- Uncommanded phase sequence, current imbalance or unexplained heating.
- Field distortion or shorted-turn signature after passive hardware installation.
- HTS temperature-margin loss or quench-protection anomaly.
- DAQ time, calibration or configuration mismatch.
- Arc, smoke, odor, coolant leak, abnormal sound or unexpected temperature rise.

Tag the discrepancy, preserve raw data, isolate affected hardware, identify configuration and obtain documented disposition before resuming.

16. Sources and controlled references

- CRTFE V0.3 Prototype Test Rig Revision 1.2.
- CRTFE V-2 Hybrid MHD Research Baseline P1.2.
- CRTFE V-2 G2 Electromagnetic Verification and Validation P1.
- CRTFE V-2 ARC and Akashic Record Vessel Integration Revision 1.4.
- CRTFE V5 Energy Sled Preliminary Engineering Manual P0.
- OSHA 29 CFR 1910.147: https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.147
- OSHA 29 CFR 1910.333: https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.333
- OSHA 29 CFR 1910.306: https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.306
- NASA-STD-8739.4: https://standards.nasa.gov/standard/NASA/NASA-STD-87394
- FAA AC 43.13-1B: https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentid/99861
- NFPA 70E: https://www.nfpa.org/codes-and-standards/nfpa-70e-standard-development/70e

References inform the safety and workmanship basis. They do not turn this preliminary project package into a code-compliant installation, flight release or certification approval.

APPENDIX A

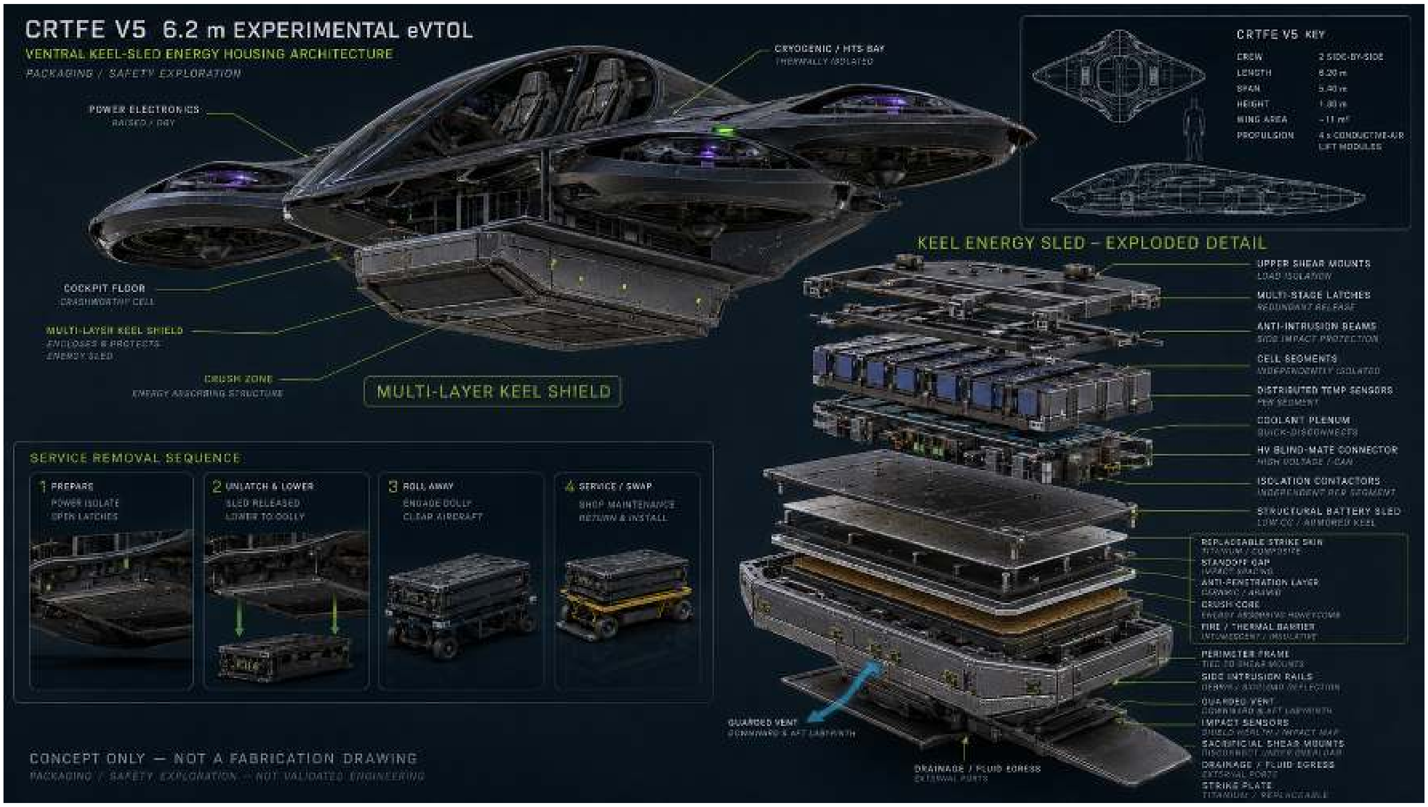
Retained V5 P0 Energy-Sled Mechanical, Shield, Fastener and Inert-Assembly Baseline

The following P0 manual pages are incorporated without changing their original claims or candidate values. P2 electrical architecture controls where the two documents overlap.

CRTFE V5

VENTRAL KEEL-SLED ENERGY HOUSING

Preliminary engineering definition, parts list, mockup build instructions, fastener schedule, calculations, and verification gates



CONCEPT ARCHITECTURE ONLY - NOT A FABRICATION DRAWING - NOT FLIGHT READY

Scope: This package defines a safe, non-energized full-scale packaging mockup and a preliminary design basis for later professional engineering. It does not authorize installation of live cells, high voltage, cryogens, pressurized coolant, powered lift hardware, tethered flight, or crewed testing.

Revision	P0 - preliminary architecture	Date	2026-08-23
Vehicle basis	CRTFE V5 target concept	Design state	Unvalidated / research

1. Design basis and hard limits

The project data support a packaging study, not a flight design. The following values are carried directly from the supplied CRTFE V5 concept and V4/V5 simulation summary. Any value not listed here is an assumption or a preliminary candidate.

Parameter	Project value	Design consequence
Crew	2, side-by-side	Energy system must remain outside protected occupant volume.
Gross mass	650 kg	No mature growth reserve in current budget.
Vehicle envelope	6.20 m L x 5.40 m span x 1.80 m H	Keel sled must preserve ground clearance and egress.
Battery-system allowance	120 kg	Includes cells, enclosure, BMS, switching, cooling, shield and wiring unless budget is revised.
Central power bay	Approx. 1.60 m longitudinal station	Sled placed at/near CG; final station requires weight-and-balance model.
Working hover input	Approx. 228 kW total	Unvalidated placeholder; drives high-power bus and severe thermal demand.
Lift architecture	4 modules	Four inverter feeds and module-level fault isolation.
Cryogenic/thermal bay	Approx. 1.10 m, aft	Separate from battery bay and occupant cell.
Design status	Conditional simulation	No energized or flight build until test gates close.

Decision gate: The 120 kg allowance and 228 kW placeholder do not by themselves define a battery. Chemistry, voltage, usable energy, continuous and pulse C-rate, thermal rejection, reserve, and thermal-runaway behavior must be selected and tested before cell count or bus voltage can be released.

Preliminary packaging envelope

A removable sled envelope of **1.45 m x 1.05 m x 0.34 m maximum** is used only to coordinate space. Gross volume is 0.518 m3. It does not define usable cell volume; shielding, crush space, cooling, buswork, switching, clearances, mounts and vent passages consume a substantial fraction.

2. Math traceability and feasibility checks

Check	Equation / input	Result	Interpretation
Weight	650 kg x 9.80665 m/s2	6.37 kN	Matches supplied hover sanity check.
Module thrust	6.37 kN / 4	1.59 kN/module	Use updated 650 kg vehicle basis, not older 450 kg simulation baseline.
Energy - 5 min	228 kW x 5/60 h	19.0 kWh	Before reserve and conversion losses.
Energy - 10 min	228 kW x 10/60 h	38.0 kWh	Before reserve and conversion losses.
Energy - 15 min	228 kW x 15/60 h	57.0 kWh	Before reserve and conversion losses.
Illustrative 120 kg system	120 kg x 200 Wh/kg system x 80% usable	19.2 kWh usable	About 5.1 min at 228 kW; 200 Wh/kg is an assumption, not project data.
Power density	228 kW / 120 kg	1.90 kW/kg system	Cells, interconnects, cooling and contactors must all support this duty.
Sled gross volume	1.45 x 1.05 x 0.34	0.518 m3	Packaging envelope only.

Mass-allocation working target (must be reconciled)

Subassembly	Working target	Notes
Cells/modules	75-85 kg	Selected only after chemistry and C-rate tests.
Structural sled + internal frames	10-14 kg	FEA and modal test required.
Multi-layer shield + crush core	10-16 kg	Likely mass driver; cannot be finalized without threat/load definition.
Cooling hardware and coolant	6-10 kg	Excludes vehicle heat exchanger if accounted elsewhere.
BMS, contactors, precharge, fusing, buswork	6-9 kg	Voltage and fault-current dependent.
Harness, sensors, seals, fasteners	3-5 kg	Includes service disconnects.
Total range	110-139 kg	Current 120 kg budget sits inside range but has little margin.

The mass range demonstrates why the concept package warned that the vehicle may move into a 750-900 kg class. Shield mass cannot be added without a full aircraft mass-budget update.

3. Blueprint-style architecture plate

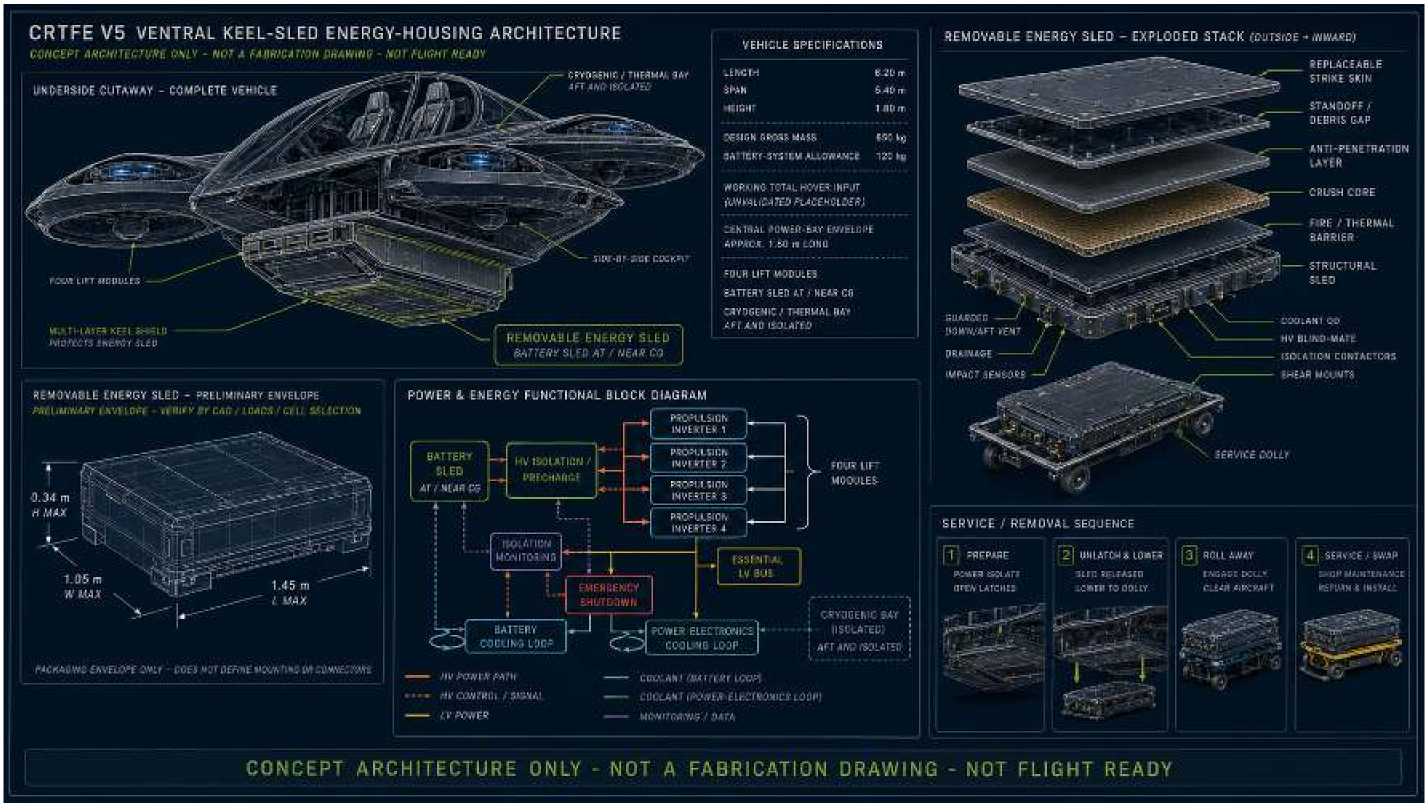


Figure 1. Packaging architecture derived from supplied project data. The 1.45 x 1.05 x 0.34 m sled is a preliminary coordination envelope only.

Interface zones

Zone	Required separation / function
Occupant cell	No cells, HV contactors, pressure relief, or vent discharge inside the crashworthy cell.
Battery sled	Segmented modules; service-disconnect state visible; independent monitoring and isolation.
Keel shield	Replaceable strike skin, standoff gap, anti-penetration layer, crush core, thermal barrier, structural sled.
Vent path	Guarded downward/aft route; must be proven by gas/particle/fire test and must not threaten occupants, controls, landing gear, or lift ducts.
Cooling	Battery and power-electronics loops segregated; leak detection and dry-break couplings at service boundary.
Cryogenic bay	Aft, thermally isolated; quench protection and energy dump remain separate development work.

4. Preliminary parts list - structure and shielding

ID	Qty	Item / candidate material	Preliminary specification / disposition
S-001	1	Structural sled frame - 7075-T73 aluminum or Ti-6Al-4V local fittings	Welded aluminum is not assumed. Use machined/extruded members with isolated bolted joints; final alloy and temper by corrosion/FEA review.
S-002	1	Inner fire/thermal pan - Inconel 625 or 321 stainless thin sheet	Thickness TBD by fire and pressure test; electrically bonded/grounded as required.
S-003	1	Microporous insulation or aerospace ceramic-fiber blanket	Chemistry-compatible, low smoke/toxicity; validate compression and moisture behavior.
S-004	1	Aluminum honeycomb crush core	Nominal 25-50 mm study range; density/crush plateau selected by impact test.
S-005	1	Anti-penetration layer - aramid laminate with optional ceramic strike tiles	Threat definition and test coupons required; do not claim ballistic rating.
S-006	1	Standoff/debris frame - 7075-T73 rails	Creates inspectable gap; includes drain paths and vent labyrinth.
S-007	1	Replaceable strike skin - Ti-6Al-4V sheet or hybrid titanium/composite	Thickness TBD after impact spectrum, ground clearance and mass trades.
S-008	2	Side intrusion rails - 7075-T73 or Ti-6Al-4V	Terminate into sacrificial load paths, not battery modules.
S-009	8	Sacrificial shear-mount cartridges	Replaceable, serialized; release load and energy absorption TBD by test.
S-010	4	Multi-stage latches with positive visual lock indication	Ground mockup may use industrial draw latches; flight candidates require qualified aerospace hardware.
S-011	1	Perimeter environmental seal - fluorosilicone	Final compound must be electrolyte/coolant/fire-agent compatible.
S-012	1	EMI/bonding gasket	Continuity target and lightning/HIRF environment TBD.
S-013	8	Impact/strain sensor nodes	Tri-axial accelerometer or piezoelectric witness; measurement range selected after drop-test plan.
S-014	1	Service dolly, braked casters, mechanical lift table	Rated at least 2x mockup mass; never work beneath unsupported sled.
S-015	4	Hard-point lifting eyes / handling fixtures	Ground handling only until proof-loaded and formally approved.

5. Preliminary parts list - electrical, thermal, sensing

ID	Qty	Item	Required function / release condition
E-001	TBD	Battery modules/cells	No procurement release until voltage, energy, C-rate, thermal-runaway propagation and supplier quality plan are approved.
E-002	1	Battery management system	Cell voltage/temperature, current, insulation state, contactor weld detection, data logging and independent shutdown path.
E-003	2	Main HV contactors	Series isolation; voltage/current/fault rating TBD. Include economizer and auxiliary state contacts.
E-004	1	Precharge contactor + resistor assembly	Sized from actual bus capacitance and target precharge time; include discharge path.
E-005	1	Main fuse or pyrotechnic disconnect	Fault-current study and coordination required before selection.
E-006	1	Manual service disconnect	Tool-controlled, interlocked, touch-safe, visible state.
E-007	1	Isolation monitoring device	Continuous HV-to-chassis monitoring; thresholds set by bus voltage and safety assessment.
E-008	1 set	HV busbars and flexible links	Laminated/insulated; ampacity, creepage, clearance and short-circuit force analysis TBD.
E-009	1	Blind-mate HV connector	Finger-safe, first-mate/last-break interlock, environmental sealing; rating TBD.
E-010	1	Low-voltage essential bus interface	Galvanically isolated from propulsion bus; supports emergency shutdown and data recorder.
T-001	1	Battery cold plate / cooling manifold	Leak-tested; electrically isolated; flow distribution verified with instrumented dummy modules.
T-002	2	Dry-break coolant quick disconnects	Keyed supply/return, low spill, pressure rating TBD.
T-003	1	Pressure relief / expansion volume	Cooling-loop component; not the battery thermal-runaway vent.
T-004	1	Guarded thermal-runaway vent labyrinth	Cross-section and burst features TBD by gas-generation and propagation tests.
T-005	2	Leak sensors	Placed at low points and connector interface.
I-001	24+	Temperature sensors	At cells/modules, coolant inlet/outlet, shield cavity and vent path.
I-002	4+	Gas/smoke sensing points	Chemistry-specific detection study required.
I-003	1	Sled data recorder / event logger	Independent timestamped capture of BMS, isolation, impact and thermal data.
W-001	1 lot	LV harness, shielded twisted pairs, backshells and clamps	Separation from HV; service loop; chafe protection; bend radius and connector keying controlled.
W-002	1 lot	Bonding straps and conductive hardware	Bonding/grounding plan and corrosion isolation required.

6. Candidate fastener schedule

These selections are detailed enough to build a non-energized fit-check mockup. They are not released flight fasteners. Critical-joint diameter, grip, preload, torque and quantity require a joint load model, bearing/bypass analysis, pull-through testing, fatigue and vibration testing, and galvanic review. Torque changes with finish, lubricant, reuse and prevailing-torque nut condition.

ID / joint	Candidate size and material	Qty	Washer / nut / lock	Mockup torque only	Flight release requirement
F-01 Strike-skin removable panels	M6 x 1.0, A286 stainless, socket flange bolt	24	A286 large-area washer; A286 all-metal prevailing-torque nut	7-9 N.m dry	Panel flutter, impact, insert pull-out and vibration test.
F-02 Standoff frame to perimeter frame	M8 x 1.25, A286 stainless hex-flange bolt	16	A286 washer + all-metal locknut	20-24 N.m dry	Combined shear/tension analysis; calibrated torque-tension test.
F-03 Anti-intrusion side rails	M8 x 1.25, Ti-6Al-4V bolt where galvanically isolated	12	Ti or isolated A286 washer/nut; insulating sleeves at carbon interfaces	TBD by coupon	Bearing/bypass, galvanic, fatigue and impact validation.
F-04 Structural sled corner joints	M10 x 1.5, A286 or alloy-steel aerospace bolt candidate	16	Hardened washer + all-metal locknut	Do not torque for flight	Preloaded-joint analysis; proof and fatigue test.
F-05 Sled-to-airframe shear mounts	M12 x 1.75 equivalent preliminary envelope; A286/Inconel candidate	8	Spherical washer pair if misalignment requires; positive secondary retention	No provisional flight torque	Aircraft loads model, crash pulse, mount release curve, fail-safe assessment.
F-06 Internal module hold-downs	M6 x 1.0 A286 with insulated bushings	24-48	Large-area washer; locking nut; captive where possible	6-8 N.m dry	Cell/module supplier limits, dielectric isolation and shock test.
F-07 Cold plate to module rails	M5 x 0.8 A286	24+	Belleville stack only if analysis supports; all-metal locknut	3.5-4.5 N.m dry	Thermal interface pressure map and cycling test.
F-08 Contactor/BMS brackets	M5 x 0.8 A286 or 316 stainless	12-20	Flat washer + all-metal locknut	3.5-4.5 N.m dry	Vibration, electrical clearance and bracket fatigue.
F-09 Service panels	M5 x 0.8 A286 captive screw	20-32	Floating nutplate; conductive or isolated finish per bonding plan	3-4 N.m dry	Repeated-removal endurance and FOD retention.
F-10 Sensor brackets / P-clamps	M4 x 0.7 A286	20+	Washer + all-metal nut or keyed nutplate	1.8-2.2 N.m dry	Harness pull, chafe and vibration test.
F-11 Grounding/bond straps	M5 x 0.8 A286	8+	Toothed bonding washer only where approved; corrosion seal after test	Per bonding test	Milliohm target, lightning/HIRF and corrosion plan.
F-12 Dolly handling fixtures	M10 x 1.5 class 10.9 zinc-nickel steel, ground equipment only	16	Class 10 all-metal nut + hardened washer	45-50 N.m dry	Not aircraft hardware; proof-load dolly separately.

Fastener control notes: use lot traceability; record grip length and installed orientation; avoid threads in primary shear planes; use metallic/composite isolation sleeves where required; use calibrated tools; record running torque separately from final torque; do not substitute nylon-insert nuts near heat/fire zones; use witness marks only as inspection aids, not as the locking method.

7. Build instructions - non-energized packaging prototype

1	**Freeze the mockup configuration** Issue a drawing index, mass target, coordinate system, preliminary CG station and keep-out zones. Mark every part MOCKUP ONLY. Use inert dimensional cell blocks; no live cells, HV, cryogens, pressurized coolant or pyrotechnic disconnects.
2	**Build a full-scale interface fixture** Construct a rigid fuselage/keel fixture representing the 1.60 m bay, cockpit floor, landing-gear keep-out, duct keep-out and service-dolly approach. Survey datum points before assembly.
3	**Fabricate the structural sled** Machine/assemble the 1.45 x 1.05 x 0.34 m maximum frame. Fit sleeves/inserts at composite interfaces. Deburr, edge-break, clean and inspect. Add temporary ballast pockets so mass and CG can be adjusted safely.
4	**Install segmented inert modules** Use dimensionally representative noncombustible blocks. Preserve proposed module gaps, cooling passages, sensor paths, HV clearance volumes and removal access. Record actual occupied volume.
5	**Install dummy electrical and thermal interfaces** Use nonconductive mock busbars, dead contactor envelopes, dead service disconnect, dummy coolant manifolds and keyed connector shells. Verify hand access, tool swing and interlock sequencing.
6	**Assemble inner containment and shield stack** Fit structural pan, thermal barrier, crush core, anti-penetration layer, standoff frame and strike skin. Use candidate F-01/F-02 fasteners. Do not adhesive-bond the prototype unless disassembly coupons are planned.
7	**Integrate vent and drainage mockups** Use smoke-safe visualization or low-pressure air only. Confirm the route exits down/aft and remains clear of cockpit, controls, landing gear, lift ducts and hot surfaces. Do not use combustion.
8	**Fit shear mounts and latches** Install instrumented or replaceable mock mount cartridges. Verify positive lock indication, secondary retention, no single-point release, and safe lowering onto the dolly.
9	**Perform service trial** With a qualified lift plan, isolate, unlatch, lower, roll away, inspect, reinstall and relock. Time each action. Confirm no person is under suspended hardware.
10	**Run fit, stiffness and handling tests** Use distributed proof loads and ballast only within a written test plan. Measure deflection, latch alignment, connector misalignment and dolly stability. Stop for cracks, yielding, binding or unexpected load transfer.
11	**Update CAD and mass properties** Weigh every subassembly, update the mass budget and envelope, and issue redlines. Do not proceed to energized hardware until the preliminary design review closes all open assumptions.
12	**Advance through gated prototypes** Coupon tests -> subscale enclosure -> instrumented inert full-scale sled -> low-energy cell module in a remote fire-test facility -> full containment test -> integrated iron-bird. Crewed installation is a separate later program.

8. Assembly manual - controls, tools and work area

Manual applicability

This procedure applies only to the full-scale inert packaging prototype. Replace all battery cells with dimensionally representative noncombustible blocks; replace HV conductors with nonconductive mock busbars; keep coolant lines dry or use low-pressure water during dedicated flow tests. A separate approved procedure is mandatory for any live cell, HV, pressurized, cryogenic, pyrotechnic, fire or propulsion work.

Control	Requirement
Minimum crew	Assembly lead plus one qualified technician. Use a third person for lifts, sled movement and independent inspection.
Document control	Use the current drawing/BOM revision. Redline deviations. Record serial/lot numbers for fasteners, inserts, latches, sensors and structural layers.
Foreign-object control	Inventory tools and loose hardware at start/end of every shift. Use covered parts bins and magnetic/nonmagnetic retrieval tools as appropriate.
Torque control	Use an in-calibration torque wrench covering the required range. Record tool ID, calibration due date, running torque, final torque and witness-mark status.
Lift safety	Use a written lift plan. Chock dolly wheels. Keep hands clear of pinch zones. Never place a person under unsupported hardware.
Material handling	Wear gloves for composites/aramid/ceramic. Control dust. Follow supplier SDS for sealants, coatings and cleaning materials.
Stop work	Stop for wrong part/revision, damaged threads, delamination, cracking, binding, misalignment, unexplained preload loss or any change to protected clearances.

Required tools and support equipment

Category	Minimum equipment
Measurement	Steel rules, digital calipers, angle gauge, feeler gauges, straightedge, laser level or survey instrument, calibrated scales.
Fastener installation	Metric hex/socket tools, open-end/ring spanners, calibrated 1-10 N.m, 5-30 N.m and 20-100 N.m torque wrenches, torque adapters, running-torque gauge.
Hole/insert work	Drill press or controlled drill guide, reamers, depth stops, countersink/deburr tools, insert installation tooling, vacuum extraction. Follow released mockup drawing.
Handling	Rated lift table/service dolly, slings/spreader where required, wheel chocks, soft jaws, nonmarring supports, temporary guide pins.
Inspection	Bright light, mirror/borescope, thread gauges, crack/damage magnification, continuity meter for bonding mockups only, camera and inspection tags.
Consumables	Lint-free wipes, approved cleaner, corrosion-isolation film/sleeves, fluorosilicone seal samples, removable thread witness lacquer, labels and serialized bags.
PPE	Eye protection, safety footwear, cut-resistant gloves, hearing protection for machining, respiratory/dust protection per material SDS.

9. Assembly manual - receiving and preassembly

A01	**Verify documents** Confirm BOM, fastener schedule, mockup drawings, revision status, inspection plan and lift plan. Write the revision on the traveler.	Builder: ____ Inspector: ____ Date: ____
A02	**Prepare the work cell** Clean the floor and benches. Establish a marked sled footprint, clean-parts table, quarantine area, FOD board and protected composite/ceramic cutting area.	Builder: ____ Inspector: ____ Date: ____
A03	**Inventory parts** Count and identify S-001 through S-015, E/T/I/W mock components and F-01 through F-12 hardware. Quarantine shortages and substitutions.	Builder: ____ Inspector: ____ Date: ____
A04	**Inspect fasteners** Reject damaged threads, mixed grades, missing material certificates, corrosion, burrs or unknown coatings. Verify size with thread gauges and grip stack with calipers.	Builder: ____ Inspector: ____ Date: ____
A05	**Inspect structural members** Check rails, corner fittings, plates and inserts for dents, cracks, warping, edge damage and blocked drain/vent features.	Builder: ____ Inspector: ____ Date: ____
A06	**Inspect shield layers** Check strike skin, standoff frame, anti-penetration panels, crush core and thermal barrier against templates. Do not install cracked ceramic or wet/damaged core.	Builder: ____ Inspector: ____ Date: ____
A07	**Prepare datum fixture** Level and secure the fuselage/keel interface fixture. Mark aircraft X/Y/Z axes, sled origin, CG reference, landing-gear keep-out and removal corridor.	Builder: ____ Inspector: ____ Date: ____
A08	**Set up the dolly** Chock wheels, verify brake function, center the lift table and install nonmarring supports. Confirm rated load is at least twice the inert mockup mass.	Builder: ____ Inspector: ____ Date: ____
A09	**Dry-fit structural frame** Assemble S-001 members with temporary guide pins. Measure diagonals, twist and envelope before installing final mockup fasteners.	Builder: ____ Inspector: ____ Date: ____
A10	**Record baseline dimensions** Record length, width, height, diagonal difference, rail parallelism and empty-frame mass. Obtain inspector acceptance before continuing.	Builder: ____ Inspector: ____ Date: ____

HOLD POINT HP-1: Do not proceed until the empty frame is dimensionally accepted, all parts are identified, and the dolly/lift setup is approved.

10. Assembly manual - structural sled and interfaces

A11	**Install permanent mockup frame fasteners** Replace guide pins one location at a time. Install F-04 candidates with correct washers and all-metal locknuts. Keep bolt threads out of primary shear planes where the grip stack permits.	Builder: ____ Inspector: ____
A12	**Snug in cross-pattern** Bring corner joints to uniform snug condition in a diagonal sequence. Measure frame diagonals again before any final mockup torque.	Builder: ____ Inspector: ____
A13	**Apply mockup torque** Apply only the schedule's mockup torque where one is stated. For F-04, do not assign a flight torque; use a documented low assembly preload selected by the assembly lead for fit testing.	Builder: ____ Inspector: ____
A14	**Install anti-intrusion rails** Fit S-008 rails using F-03 candidates, isolation sleeves and large bearing washers. Verify no hard point can contact an inert module block under nominal assembly tolerance.	Builder: ____ Inspector: ____
A15	**Install inner fire/thermal pan** Place S-002 without wrinkles or sharp unsupported edges. Keep drains and vent entrances open. Bonding jumpers remain mock interfaces unless a bonding test plan is active.	Builder: ____ Inspector: ____
A16	**Fit thermal barrier** Install S-003 with full coverage and controlled seams. Avoid compression beyond the supplier's provisional limit. Photograph seams before they are hidden.	Builder: ____ Inspector: ____
A17	**Install dummy cold plate** Fit T-001 with F-07 hardware. Use thickness-controlled dummy interface pads. Verify flatness and that coolant ports face the service boundary.	Builder: ____ Inspector: ____
A18	**Install inert module blocks** Place numbered inert blocks in the planned segmentation pattern. Install F-06 hold-downs hand-snug, then use the provisional mockup torque. Record each block mass and position.	Builder: ____ Inspector: ____
A19	**Install dummy HV/LV components** Fit E-002 through E-010 dead envelopes and nonconductive busbars. Preserve service clearances, connector keying and emergency-disconnect access.	Builder: ____ Inspector: ____
A20	**Install mock harness and sensors** Route W-001/W-002 mock harnesses and I-001 through I-003 sensor bodies. Maintain separation, bend radius, clamp spacing and chafe protection. Tag both ends.	Builder: ____ Inspector: ____
A21	**Install dry coolant interfaces** Fit T-002/T-003 dry-break envelope models and dry hoses. Verify supply/return keying and ensure no hose crosses the removal path or vent outlet.	Builder: ____ Inspector: ____
A22	**Intermediate inspection** Inspect concealed fasteners, module restraints, harness routing, thermal barrier coverage, drainage and connector access. Record photographs and discrepancies.	Builder: ____ Inspector: ____

HOLD POINT HP-2: Independent inspection required before closing the sled or installing shield layers.

11. Assembly manual - shield stack, mounts and installation

A23	**Fit perimeter frame and seal** Install S-011 seal sample without stretching. Verify corners, joint gap and drainage discontinuities match the mockup drawing.	Builder: ____ Inspector: ____
A24	**Install crush core** Fit S-004 within the perimeter frame. Do not crush during installation. Measure thickness at the defined grid and record any local low spots.	Builder: ____ Inspector: ____
A25	**Install anti-penetration layer** Place S-005 over the crush core using designated edge supports. If tiled, stagger seams and record tile map/serials. Keep the guarded vent corridor open.	Builder: ____ Inspector: ____
A26	**Install standoff/debris frame** Fit S-006 using F-02 candidates. Snug in a cross-pattern, verify standoff gap, then apply 20-24 N.m dry mockup torque only after measuring running torque.	Builder: ____ Inspector: ____
A27	**Install strike skin** Place S-007 using lifting aids if required. Start all F-01 bolts by hand. Snug center-out in a cross-pattern, verify flushness, then apply 7-9 N.m dry mockup torque.	Builder: ____ Inspector: ____
A28	**Install impact sensors** Mount S-013/I-001 sensor bodies and protected leads. Verify connectors cannot be crushed during shield removal.	Builder: ____ Inspector: ____
A29	**Verify vent and drainage** Use low-pressure air and smoke-safe visualization only. Confirm flow exits down/aft and no blocked drain or direct line into occupant/gear/duct zones exists.	Builder: ____ Inspector: ____
A30	**Install shear-mount cartridges** Fit S-009 mock cartridges to the fixture. Use temporary alignment pins, then install F-05 candidates without assigning flight torque. Apply only the test-plan assembly condition.	Builder: ____ Inspector: ____
A31	**Install multi-stage latches** Fit S-010 latches and secondary retention. Adjust evenly so the sled seats without frame distortion. Confirm visible locked/unlocked state.	Builder: ____ Inspector: ____
A32	**Position sled on dolly** Center the sled and secure it to the dolly handling fixture. Confirm CG is within the dolly support polygon before movement.	Builder: ____ Inspector: ____
A33	**Raise sled into bay** Use the lift table in small increments with one lead directing. Stop at any interference. Use guide pins; never use fasteners or latches to pull a misaligned sled into place.	Builder: ____ Inspector: ____
A34	**Engage mounts and latches** Seat all mount interfaces, remove guide pins, close primary latches, then engage secondary retention. Confirm no single latch carries the assembly.	Builder: ____ Inspector: ____
A35	**Connect dummy interfaces** Mate blind-mate connector envelopes and dry coolant QDs by hand. Verify full engagement indicators and no forced alignment.	Builder: ____ Inspector: ____
A36	**Final torque/locking inspection** Audit installed fasteners against F-01 through F-12, record tool IDs and values, apply witness marks where specified, and complete FOD inventory.	Builder: ____ Inspector: ____

A37	**Final dimensional survey** Record installed envelope, ground clearance, gear/duct/control clearances, sled-to-bay gaps and shield alignment.	Builder: ____ Inspector: ____
A38	**Mass and CG update** Weigh the complete inert sled and shield, calculate its CG, update vehicle mass properties and compare with the 120 kg system allowance.	Builder: ____ Inspector: ____
A39	**Release for inert tests** Assembly lead and independent inspector sign the traveler. Release only for the approved inert fit, handling, modal, airflow or low-pressure tests.	Builder: ____ Inspector: ____

HOLD POINT HP-3: The completed inert assembly may not be energized. Any live-cell or HV transition requires a new released configuration, hazard analysis, remote test plan and professional approval.

12. Assembly manual - removal, reinstallation and inspection

R01	**Establish safe state** For the inert prototype, verify all mock services are disconnected. For any future energized article, use a separate lockout/tagout and zero-energy verification procedure.	Tech: ____ Inspector: ____
R02	**Position and chock dolly** Center the lift table below the sled, apply wheel brakes and install lateral guides.	Tech: ____ Inspector: ____
R03	**Support sled** Raise the dolly until supports contact designated pads. Confirm stable four-point support before touching latches.	Tech: ____ Inspector: ____
R04	**Disconnect dummy interfaces** Unmate dry coolant and connector envelopes by the intended release features. Cap/protect interfaces.	Tech: ____ Inspector: ____
R05	**Release secondary retention** Remove safety clips/pins in the defined order and account for each item.	Tech: ____ Inspector: ____
R06	**Open primary latches** Open latches evenly. If binding occurs, stop and correct support height; do not pry or hammer.	Tech: ____ Inspector: ____
R07	**Disengage mount interfaces** Install guide pins as required, release mock shear-mount retention and verify the sled is fully supported.	Tech: ____ Inspector: ____
R08	**Lower in controlled increments** Lower while checking all sides. Stop for interference, harness snagging or unexpected tilt.	Tech: ____ Inspector: ____
R09	**Roll to service area** Lower to transport height, verify CG and restraints, release brakes only when the path is clear, then move at walking speed.	Tech: ____ Inspector: ____
R10	**Remove strike skin** Support S-007. Remove F-01 bolts in an outside-to-center cross-pattern. Bag/tag hardware by location.	Tech: ____ Inspector: ____
R11	**Inspect shield stack** Inspect strike skin dents, standoff deformation, core crushing, aramid/ceramic damage, heat/fire-barrier condition, vent blockage and sensor witness data.	Tech: ____ Inspector: ____
R12	**Inspect sled** Inspect mounts, latches, fasteners, frame, inert module restraints, connectors, harnesses, cold plate and drainage. Record damage with scale and location.	Tech: ____ Inspector: ____
R13	**Disposition discrepancies** Use accept-as-is, repair, replace or engineering review dispositions. Do not reinstall damaged critical parts without written approval.	Tech: ____ Inspector: ____
R14	**Reassemble and reinstall** Repeat A23-A39. Replace prevailing-torque nuts when required by the selected hardware specification or when running torque is out of limit.	Tech: ____ Inspector: ____

13. Assembly traveler and sign-off checklist

Record	Entry
Assembly serial / configuration	____
Drawing/BOM revision	____
Work order / test article	____
Assembly lead / technician	____
Independent inspector	____
Torque wrench IDs and calibration dates	____
F-01/F-02/F-03/F-04/F-05 lots	____
Shield layer serials / material lots	____
Measured empty sled mass and CG	____
Measured complete sled mass and CG	____
Installed clearances	Ground ____ Gear ____ Duct ____ Cockpit ____
HP-1 accepted by/date	____
HP-2 accepted by/date	____
HP-3 accepted by/date	____
Open discrepancies / NCRs	____
Released inert test scope	____

Final check	Pass / fail / N/A	Inspector initials
No live cells, HV, cryogens, pressure or pyrotechnics installed	________	________
Fastener IDs, grip, locking and mockup torque verified	________	________
All concealed-work photographs attached	________	________
FOD/tool inventory complete	________	________
Vent and drainage paths open	________	________
Mounts and latches fully engaged with secondary retention	________	________
Ground/gear/duct/cockpit clearances recorded	________	________
Dolly and handling fixtures removed or secured	________	________
Mass properties updated	________	________
Test limitations placarded	________	________

14. Inspection plan and acceptance gates

Gate	Method	Preliminary pass condition
Dimensional inspection	CMM/laser tracker or surveyed fixture	Envelope, datums and service clearances match released mockup drawing; deviations recorded.
Mass properties	Calibrated scales and balance fixture	Measured mass and CG entered into aircraft budget; no hidden growth.
Fastener installation	100% traveler review	Correct ID, lot, grip, washer, locking method, torque-tool ID and inspector signoff.
Latch integrity	Repeated cycling and proof pull	Positive lock indication after service cycles; no binding or permanent deformation.
Shield impact coupons	Drop/impactor coupons	No unacceptable intrusion into protected-volume witness block; criteria set before test.
Crush-core response	Quasi-static crush coupons	Stable plateau and energy absorption within selected tolerance.
Vent path	Cold-flow/smoke visualization, then remote fire facility	No cockpit/gear/duct impingement; pressure and temperature remain within derived limits.
Water/dust ingress	Spray/dust test on empty enclosure	No pooling at cells/connectors; drains work; seals remain inspectable.
Modal/vibration	Instrumented shaker on inert sled	No latch release, fastener back-out, fretting, wire chafe or harmful resonance.
Electrical safety	Low-voltage continuity/insulation mock tests first	Interlocks and emergency shutdown logic fail safe; energized testing requires separate hazard review.
Thermal system	Flow bench with heaters and inert modules	Flow balance, leak rate, pressure drop and heat rejection meet requirements derived from component tests.
Containment	Remote destructive test with representative cells	No hazardous propagation or discharge into protected zones; criteria approved before test.

Stop-work triggers: unexpected heating, odor/gas, electrical continuity in an intended isolated zone, cracking/delamination, fastener rotation, latch migration, coolant leakage, unstable dolly motion, or any test result outside the pre-approved envelope.

15. R&D; company work products still required

Discipline	Required controlled deliverables before flight release
Requirements	System requirements, interface-control document, hazard analysis, compliance basis, verification cross-reference matrix.
Aerostructures	3D CAD, drawing tree, material allowables, laminate schedules, loads model, FEA, fatigue/damage-tolerance, modal/flutter integration.
Crashworthiness	Crash pulse definition, mount load paths, intrusion limits, occupant-cell interaction, dynamic tests and post-crash egress assessment.
Battery	Chemistry and supplier qualification, cell/module characterization, energy/power sizing, abuse/propagation testing, BMS safety requirements.
Electrical	Voltage architecture, short-circuit/fault-current model, HVIL, creepage/clearance, insulation coordination, grounding/bonding, EMC/HIRF/lightning.
Thermal/fire	Heat-load model, cooling sizing, vent gas characterization, pressure analysis, fire detection/suppression, drainage and materials flammability.
Fasteners/joints	Joint-by-joint load cases, grip stack, preload scatter, torque-tension tests, composite inserts, galvanic isolation, locking and inspection plan.
Maintainability	Removal tooling, lift plan, lockout/tagout, service intervals, connector cycles, damage limits and illustrated parts catalog.
Test	Coupon, subcomponent, sled, iron-bird, environmental, vibration, impact, abuse and containment procedures with calibrated instrumentation.
Quality/configuration	Part numbering, serialized critical parts, approved suppliers, material certs, nonconformance control, drawing revision and software/data configuration.
Certification/safety review	Independent DER/ODA or appropriate certification specialists; FAA engagement strategy; operational limitations; emergency response plan.

Minimum team for the next design phase

Aerospace structures engineer; battery systems engineer; high-voltage power engineer; thermal/fire specialist; crashworthiness engineer; mechanical designer; test engineer; safety/reliability engineer; manufacturing/quality engineer; and certification counsel. No single conceptual drawing can replace these analyses.

16. THE ARC integrated command architecture

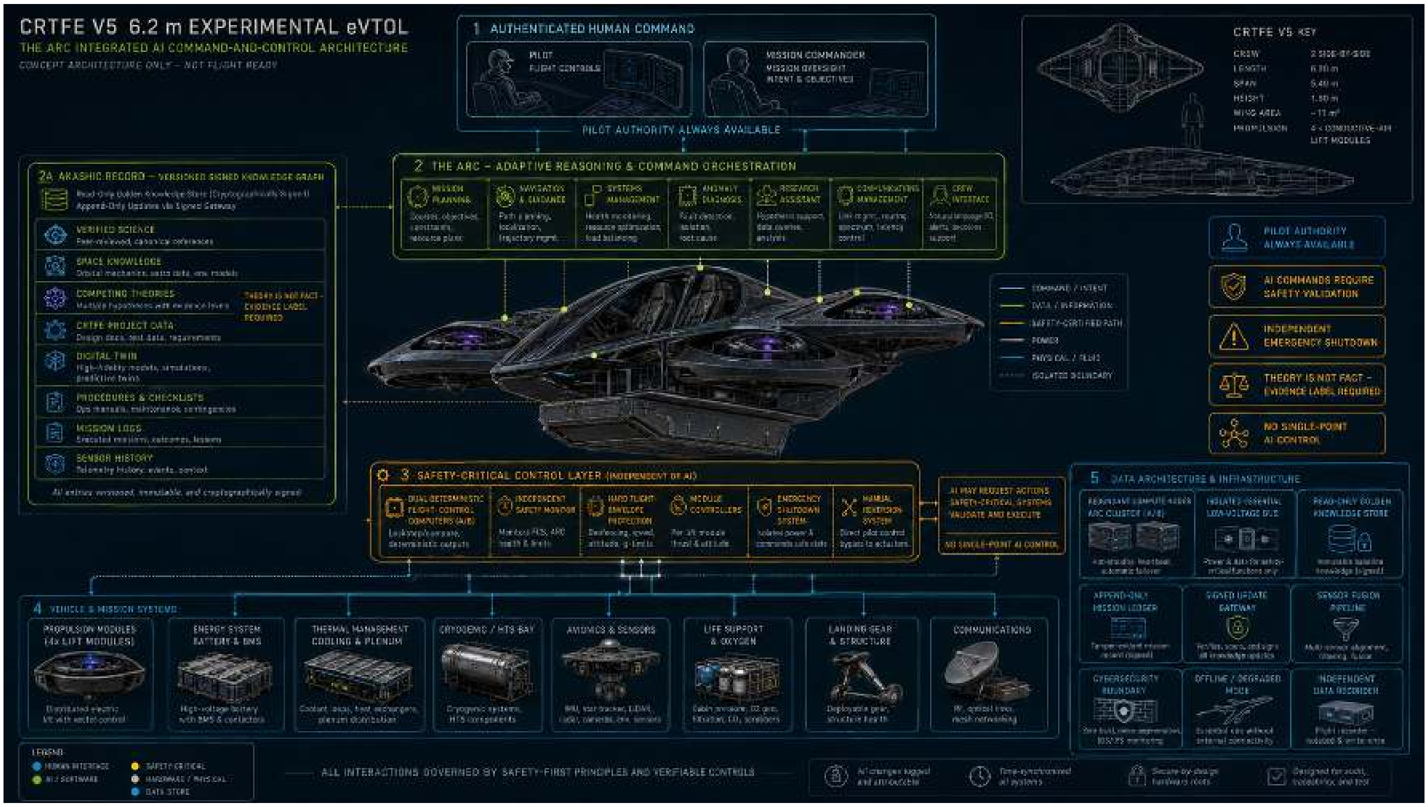


Figure 2. THE ARC vessel-wide command, knowledge and safety architecture. Concept only; not flight-ready software or avionics.

Purpose and authority

THE ARC is the vessel-wide adaptive reasoning and command-orchestration layer. Under authenticated pilot or mission-commander intent, it may plan, recommend, coordinate, diagnose and request actions across the vehicle. It is deliberately prevented from directly driving safety-critical actuators. Dual deterministic flight-control computers, module controllers, an independent safety monitor, hard flight-envelope protection and emergency shutdown validate and execute commands. Pilot authority and manual reversion remain available.

Priority	Authority source	Permitted role
1	Independent emergency shutdown	Remove propulsion energy and isolate hazardous systems when protected limits are exceeded.
2	Pilot/manual reversion	Direct validated control path independent of THE ARC.
3	Deterministic safety/flight-control layer	Execute control laws, enforce envelopes and reject unsafe or malformed requests.
4	THE ARC	Interpret human intent, optimize mission/system plans and issue bounded requests with rationale and confidence.
5	Akashic Record	Provide signed knowledge, procedures, models, evidence labels and mission context; never command hardware.

No single-point AI control: loss, corruption or disagreement of THE ARC must leave the deterministic control layer capable of continued safe operation or a predefined safe state.

17. Akashic Record knowledge system and ARC bill of materials

Akashic Record definition

The Akashic Record is a technical, cryptographically signed and versioned knowledge architecture - not a claim of supernatural access or omniscience. It stores what humanity and the project currently know, what remains disputed, and why. Every item carries provenance, date, author/source, evidence class, confidence, dependencies, contradictions, applicable vehicle configuration and validation status.

Collection	Required content / controls
Verified science	Peer-reviewed and canonical references; equations, units, applicability limits, replications and superseded versions.
Space knowledge	Orbital mechanics, astrodynamics, environments, navigation, astronomy, planetary science, communications and human factors.
Competing theories	Hypotheses stored side by side with evidence-for, evidence-against, falsification tests and confidence. Theory never promoted to fact without review.
CRTFE project	Requirements, simulations, measured tests, CAD/digital twin, BOMs, drawings, hazards, procedures and configuration history.
Mission memory	Append-only commands, sensor history, decisions, anomalies, outcomes, crew notes and lessons learned.
Procedures	Normal, abnormal and emergency checklists with approvals, effective configuration and expiration/review dates.
Model registry	AI/model versions, training/evaluation scope, known limitations, safety cases, allowed tools and rollback images.

Preliminary ARC hardware/software BOM

ID	Qty	Candidate item	Required characteristic
ARC-H01	2	Rugged redundant compute nodes A/B	Dissimilar failure analysis; hot-standby/lockstep strategy; ECC memory; secure boot; independent power feeds.
ARC-H02	1	Independent safety-monitor computer	Separate hardware/software lineage; monitors control requests, health, timing and envelopes.
ARC-H03	2	Read-only golden knowledge stores	Cryptographically signed baseline; physically or logically write-protected in flight.
ARC-H04	2+	Append-only mission ledger storage	Tamper-evident, time-synchronized, fault-tolerant and removable for investigation.
ARC-H05	1	Signed update gateway	Ground-authorized ingestion, malware scan, provenance validation, staging and rollback.
ARC-H06	2	Time/network gateways	Redundant deterministic data buses plus isolated research/crew network.
ARC-H07	1	Independent data recorder	Write-once or protected partition; separate power and crash-survivability study.
ARC-H08	2	Crew interface displays/controls	Clear mode/authority indication, confidence/provenance view, explicit confirmation for consequential commands.
ARC-H09	1 set	Cybersecurity boundary devices	Segmentation, allowlists, rate limiting, intrusion detection and removable-media control.
ARC-S01	1	Command-orchestration software	Typed/bounded command API; rationale; timeout; rollback; no raw actuator access.
ARC-S02	1	Knowledge graph + evidence engine	Provenance, evidence class, contradictions, units, configuration applicability and signed revisions.
ARC-S03	1	Digital twin / state estimator	Real-time vehicle configuration, mass properties, health and predicted margins.
ARC-S04	1	Mission planner and navigator	Constraint-based plans independently checked by certified navigation/control functions.
ARC-S05	1	Anomaly diagnosis engine	Ranked hypotheses, evidence trace, uncertainty, safe troubleshooting actions and escalation.
ARC-S06	1	Offline degraded-mode image	Essential local knowledge/procedures and deterministic interfaces without external connectivity.

18. THE ARC development and verification gates

Gate	Required demonstration before advancement
ARC-0 Knowledge prototype	Ingest project sources; preserve citations; label fact/theory/assumption; retrieve correct configuration-specific procedures; reject unsupported claims.
ARC-1 Desktop digital twin	Operate on simulated vehicle telemetry; produce bounded plans and explain every recommendation with evidence and uncertainty.
ARC-2 Software-in-the-loop	Communicate only through typed command requests; safety monitor rejects out-of-envelope, stale, malformed, contradictory and unauthorized requests.
ARC-3 Hardware-in-the-loop	Run redundant compute, buses, displays and data recorder against an iron-bird. Demonstrate timing, failover, rollback, degraded mode and loss of connectivity.
ARC-4 Fault injection	Corrupt sensors, models, knowledge entries, timing, networks and one compute node. Show safe detection, isolation and continued deterministic control.
ARC-5 Human factors	Pilots can understand authority, uncertainty and rationale; override is immediate; alerts do not overload or mislead; mode confusion is controlled.
ARC-6 Cybersecurity	Independent threat model, penetration testing, signed update validation, key management, supply-chain review and recovery from compromised noncritical nodes.
ARC-7 Ground iron-bird	Coordinate energy, thermal, propulsion and avionics mock systems without direct actuator authority. Record every request and outcome.
ARC-8 Uncrewed bounded test	Only after propulsion/airframe gates, independent safety review and a flight-test authorization. Start with narrow functions and hard geofenced limits.
ARC-9 Crewed consideration	Requires a formal safety case, certification strategy, demonstrated manual reversion and extensive operational evidence. Not authorized by this package.

THE ARC is a research and systems-management concept. It must not be described as knowing everything, predicting unknown physics with certainty, or replacing certified flight controls, engineering judgment or crew authority.

19. Source basis and standards-to-review

Project source. CRFTE V5 Target Vehicle - Concept Definition Package (3 pages). Supplies the 650 kg target, geometry, 120 kg battery allocation, 1.60 m central power bay, 228 kW placeholder load, cryogenic/thermal segregation, and development gates.

Project source. CRTFE V4-V5 Simulation Summary (4 pages). Establishes the conditional nature of the propulsion model, older 450 kg/1.104 kN-per-module baseline, 60 kW-per-module cap, and unresolved conductivity/thermal questions.

FAA AC 20-184. Guidance on testing and installation of rechargeable lithium batteries and battery systems on aircraft. https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentid/1027106

FAA lithium battery resources. FAA material identifies thermal runaway, internal short circuit, overcharge, over-discharge, heat and pressure hazards. https://www.faa.gov/aircraft/air_cert/design_approvals/dah/lithium_batteries

eCFR 14 CFR Part 25. Emergency landing and retention-of-mass concepts are useful design references; the actual certification basis for this vehicle is not established. https://www.ecfr.gov/current/title-14/chapter-I/subchapter-C/part-25

NASA NTRS. Preloaded joint analysis methodology for space-flight systems provides bolted-joint analysis background. https://ntrs.nasa.gov/citations/19960012183

NASA NTRS. Aerospace threaded fastener strength in combined shear and tension highlights the need to analyze combined loading. https://ntrs.nasa.gov/api/citations/20120003667/downloads/20120003667.pdf

Standards candidates to procure/review with qualified engineers: RTCA DO-311A (rechargeable lithium battery systems); applicable SAE/NAS/MS fastener specifications; environmental qualification appropriate to the system; wiring, bonding, flammability and structural standards selected through the formal certification basis. Listing a standard here is not a claim of compliance.

Release statement

This P0 package is suitable for design discussion, budgeting, supplier RFIs, CAD coordination and construction of an inert packaging mockup. It is not suitable for procurement of live battery cells, release of flight hardware, structural proof, pressure/fire certification, energized system integration, or any crewed or uncrewed flight test.

ELECTRICAL DRAWING INDEX AND CONTROL BOUNDARY

P2 ground-research package

SHEET	DRAWING	PURPOSE
E-101	V0.3 SYSTEM INTERCONNECT	Airflow, enclosed plasma subsystem, instruments, DAQ and safety
E-102	V0.3 HARDWIRED SAFETY CHAIN	Dual-channel concept, manual reset and monitored contactor feedback
E-201	G2B STATOR VALIDATION ONE-LINE	Polyphase source, phase measurement, field mapping and force stand
E-301	G2D HTS / STATOR INTEGRATION	Independent HTS protection and traveling-wave source
E-401	V5 ENERGY-SLED SINGLE-LINE	HV functional order, BMS, IMD, branch isolation and LV boundary
E-501	ARC POWER / DATA / SAFETY	Reasoning, mediation and independent protective paths
E-601	HARNESS ZONES / INTERFACES	Cable classes, routing separation, connector IDs and hold points

P2 RELEASES

Functional connections and independent safety boundaries
Equipment and connector identifiers
Cable-service classes and separation rules
De-energized assembly and inspection sequence

P2 DOES NOT RELEASE

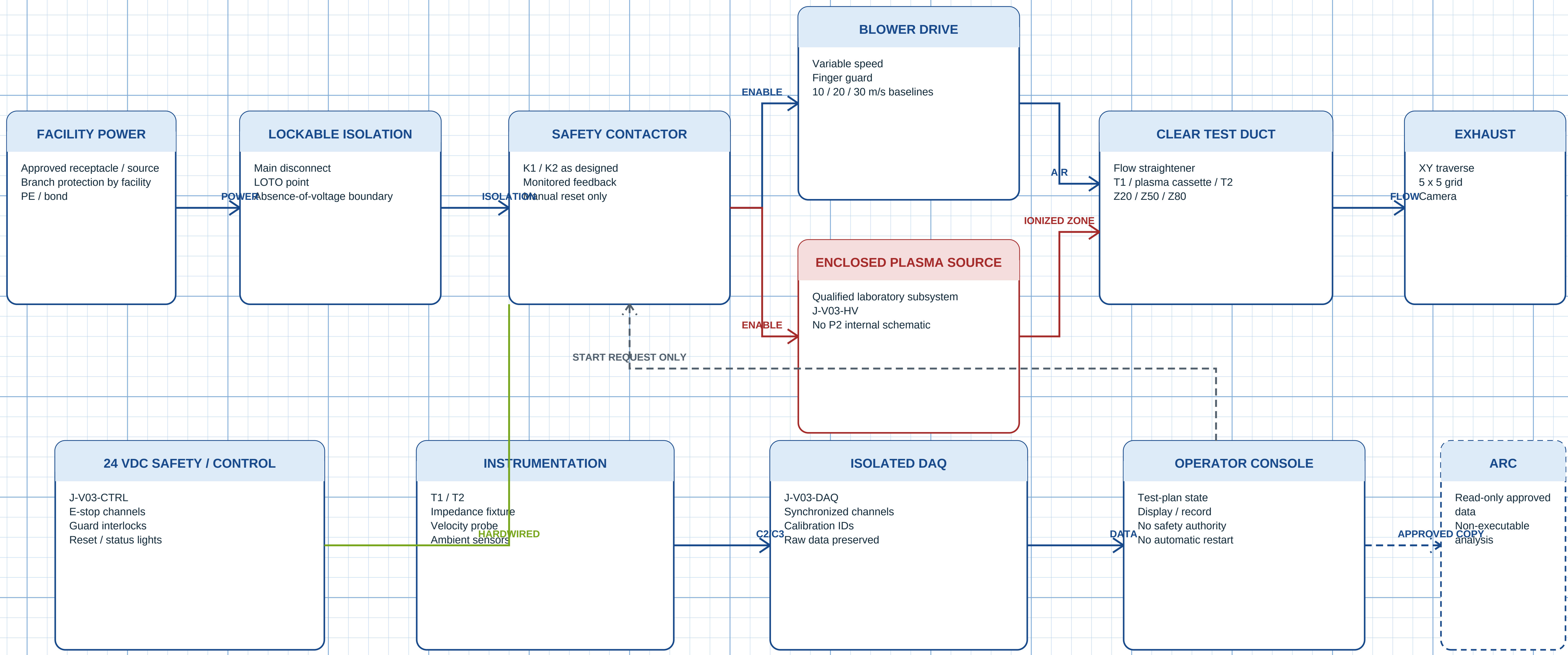
Bus voltage or hazardous-energy limits
Power-wire gauge, fuse, contactor or coil rating
Plasma-pulser construction or energized procedure
Flight hardware, fabrication or certification approval

CORE RULE: ARC, DAQ AND ORDINARY SOFTWARE NEVER CARRY THE HARDWIRED EMERGENCY-STOP OR PROTECTIVE-TRIP FUNCTION.

ELECTRICAL DRAWING INDEX AND CONTROL BOUNDARY

V0.3 MOVING-AIR TEST RIG - SYSTEM INTERCONNECT

100 x 100 mm duct | 10 / 20 / 30 m/s



NO PLASMA-ON TEST UNTIL THE ENCLOSED SOURCE, INTERLOCKS, LOTO, INSTRUMENTATION AND HOST-LAB PROCEDURE PASS THE READINESS REVIEW.

V0.3 MOVING-AIR TEST RIG - SYSTEM INTERCONNECT

UNITS: functional connection architecture | values marked TBD require engineering release

SHEET E-101

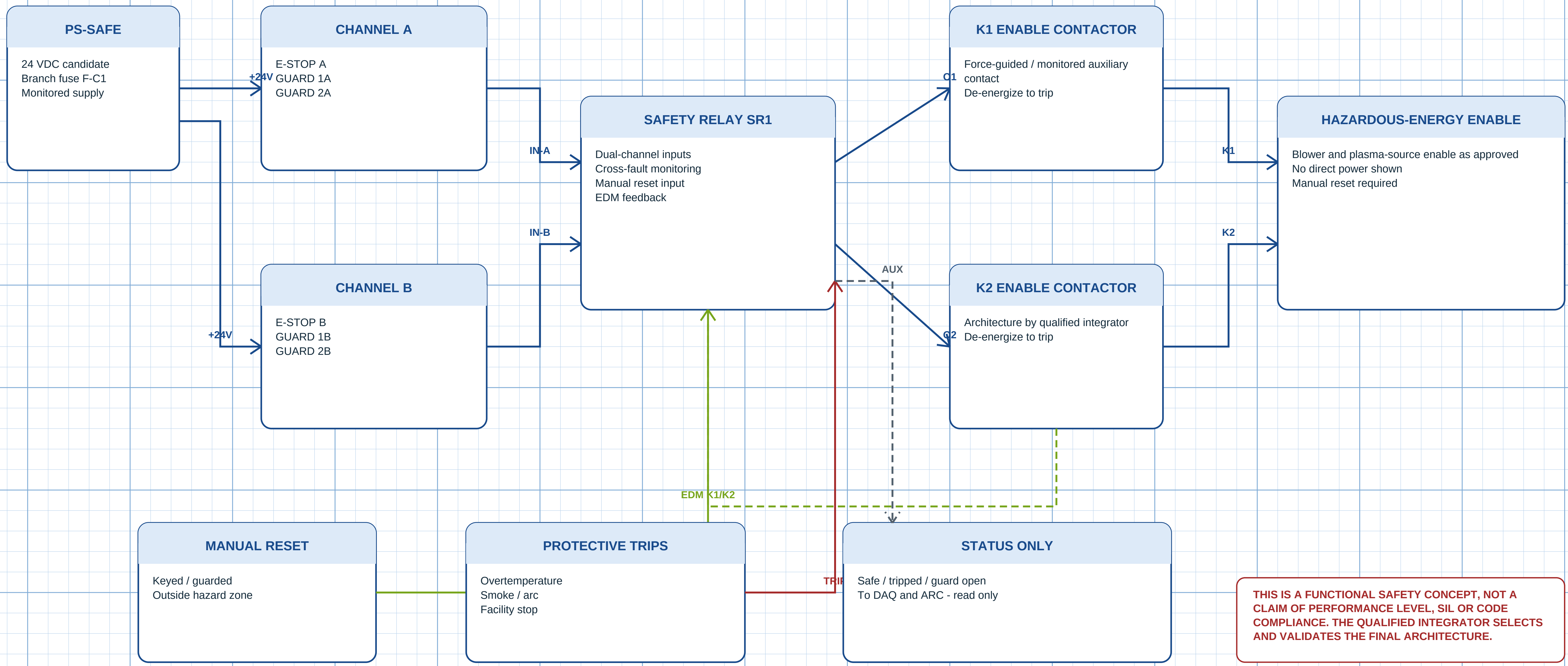
REV P2 | 2026-08-24

CRTFE V-2

CONCEPT ONLY

V0.3 HARDWIRED SAFETY CHAIN - CONCEPT

24 VDC low-energy logic shown; component category/rating TBD



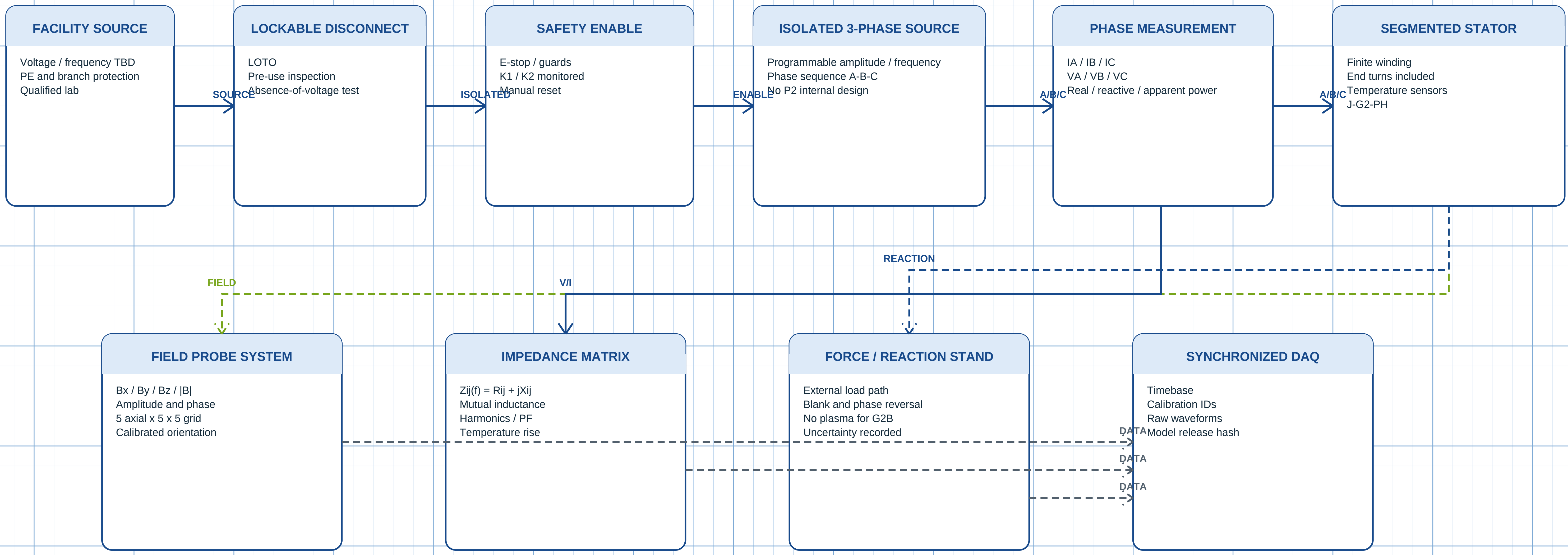
V0.3 HARDWIRED SAFETY CHAIN - CONCEPT

UNITS: functional connection architecture | values marked TBD require engineering release

SHEET E-102	CRTFE V-2
REV P2 | 2026-08-24	CONCEPT ONLY

G2B PLASMA-OFF POLYPHASE STATOR VALIDATION - ONE-LINE

125-point vector field map minimum



PASS REQUIRES MODEL-TO-MEASUREMENT AGREEMENT AGAINST PRE-REGISTERED UNCERTAINTY-BASED LIMITS. COMMANDED CURRENT OR AN IDEAL FIELD PLOT IS NOT VALIDATION.

G2B PLASMA-OFF POLYPHASE STATOR VALIDATION - ONE-LINE

UNITS: functional connection architecture | values marked TBD require engineering release

SHEET E-201

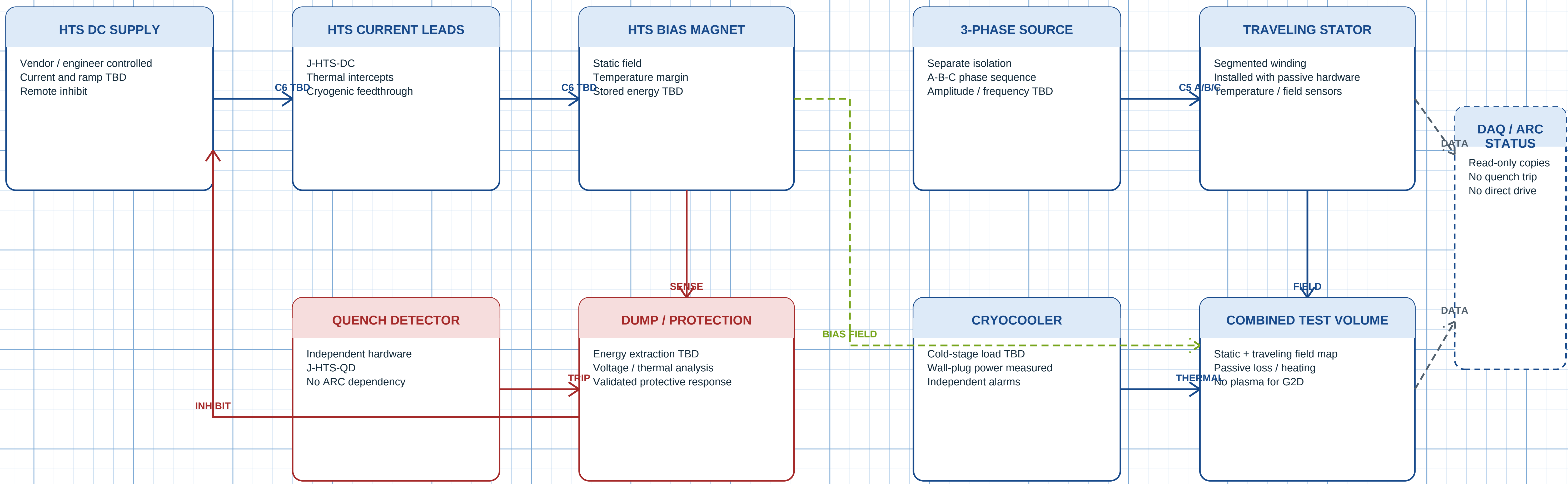
REV P2 | 2026-08-24

CRTFE V-2

CONCEPT ONLY

G2D COMBINED HTS BIAS AND TRAVELING-WAVE STATOR

plasma off | protection independent



STOP ON UNEXPLAINED HEATING, AC LOSS, FIELD DISTORTION, LOSS OF HTS TEMPERATURE MARGIN OR ANY QUENCH-PROTECTION ANOMALY.

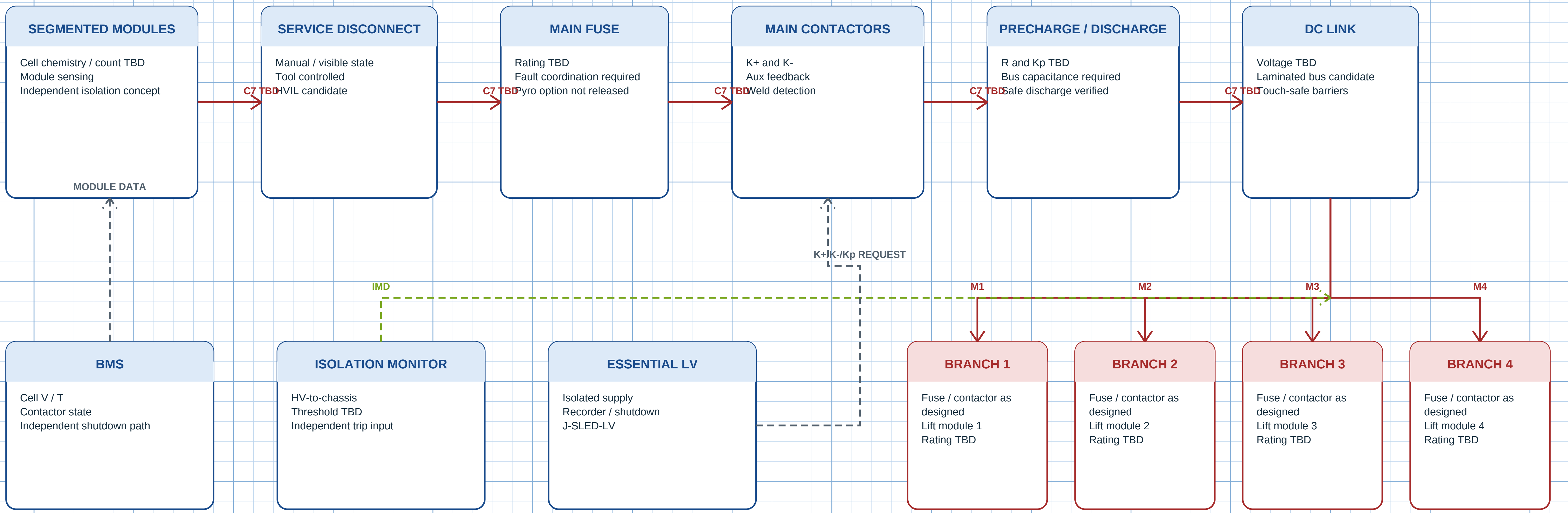
G2D COMBINED HTS BIAS AND TRAVELING-WAVE STATOR

UNITS: functional connection architecture | values marked TBD require engineering release

SHEET E-301	**CRTFE V-2**
REV P2 | 2026-08-24	CONCEPT ONLY

V5 TARGET ENERGY SLED - FUNCTIONAL SINGLE-LINE

1450 x 1050 x 340 mm envelope | 120 kg allowance



HARDWIRED PROTECTION

BMS / IMD / contactor / fuse / service disconnect remain outside ARC authority; exact circuit and ratings TBD.

CONNECTOR BOUNDARY

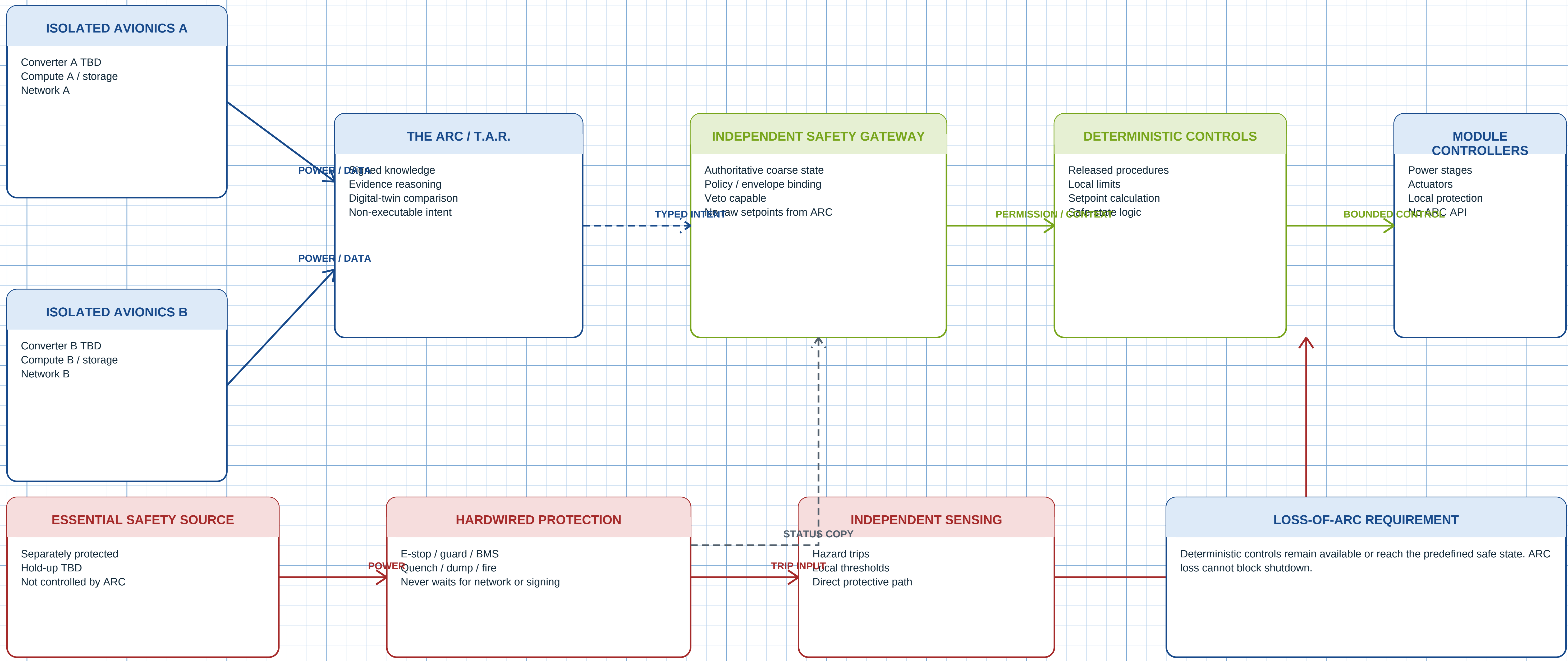
J-SLED-HV touch-safe/keyed/HVIL candidate; J-SLED-LV carries status and shutdown request only. Pinouts are not released.

NO LIVE-CELL OR HV BUILD IS AUTHORIZED BY THIS DRAWING. A SELECTED CHEMISTRY, BUS, FAULT STUDY, THERMAL PROPAGATION TEST AND RESPONSIBLE-ENGINEER RELEASE ARE REQUIRED.

V5 TARGET ENERGY SLED - FUNCTIONAL SINGLE-LINE	**SHEET E-401**	**CRTFE V-2**
UNITS: functional connection architecture | values marked TBD require engineering release	REV P2 | 2026-08-24	CONCEPT ONLY

ARC / T.A.R. POWER, DATA AND SAFETY SEPARATION

ARC Revision 1.4 | all plans non-executable



ARC / T.A.R. POWER, DATA AND SAFETY SEPARATION

UNITS: functional connection architecture | values marked TBD require engineering release

SHEET E-501	CRTFE V-2
REV P2 | 2026-08-24	CONCEPT ONLY

HARNESS ZONES, CABLE CLASSES AND RELEASE EVIDENCE

separate routing | controlled crossings | no inferred ratings

ZONE A - C0 / C1

Protective bond and 24 V safety/control
Rating / spacing by released design

ZONE B - C2 / C3

Analog sensors and thermocouples
Rating / spacing by released design

ZONE C - C4

Data/time; fiber preferred across noisy boundaries
Rating / spacing by released design

ZONE D - C5

Polyphase stator power; symmetric routing
Rating / spacing by released design

ZONE E - C6

HTS DC/current leads and quench wiring
Rating / spacing by released design

ZONE F - C7 / C8

Propulsion HV and special disconnect circuits
Rating / spacing by released design

CONTROLLED CROSSING RULE

Cross power and sensor/data harnesses only at documented locations, preferably near 90 degrees. Maintain the released separation, shielding, bonding and enclosure-feedthrough design. Do not bundle unlike classes for convenience.

INTERFACE	SERVICE	MANDATORY FEATURES	RELEASE EVIDENCE
J-V03-CTRL	V0.3 low-voltage control	Keyed; status/control only	Netlist + low-energy test
J-V03-DAQ	Sensors / DAQ	Shielded pairs; no hazardous energy	Calibration + channel map
J-V03-HV	Enclosed plasma source	Lab-owned; touch-safe; interlocked	Facility approval
J-G2-PH	Stator A/B/C	Phase ID; symmetric routing	Impedance / thermal analysis
J-HTS-DC/QD	HTS current / quench	Independent protection	Magnet safety review
J-SLED-HV/LV	Energy sled	Keyed; HVIL candidate; LV segregated	Fault + insulation + FMEA
J-ARC-A/B	ARC redundant data/power	Isolated; no actuator pins	Interface control review

WIRE SIZE, OVERCURRENT PROTECTION, CONTACTOR RATINGS, CREEPAGE/CLEARANCE AND CONNECTOR PINOUTS REMAIN TBD UNTIL HARDWARE AND ELECTRICAL LOADS ARE SELECTED.

HARNESS ZONES, CABLE CLASSES AND RELEASE EVIDENCE

UNITS: functional connection architecture | values marked TBD require engineering release

SHEET E-601

REV P2 | 2026-08-24

CRTFE V-2

CONCEPT ONLY